Homework 4

Problem 1

Noether theorem: if the Lagrangian of the system is invariant under transformation

$$q_i \to Q_i(q_i, \xi) \quad \text{where} \quad Q_i(q_i, \xi = 0) \equiv q_i$$

then

$$\sum_i \frac{\partial L}{\partial \dot{q}_i} \frac{\partial Q_i}{\partial \xi}\bigg|_{\xi=0} = \text{const}$$

(Derivation by considering $\frac{dL}{d\xi} = \sum_i \left(\frac{\partial L}{\partial Q_i} \frac{\partial Q_i}{\partial \xi} + \frac{\partial L}{\partial \dot{Q}_i} \frac{\partial \dot{Q}_i}{\partial \xi} \right)$ and then transforming it)

in our case the continuous transformation leaving L invariant is

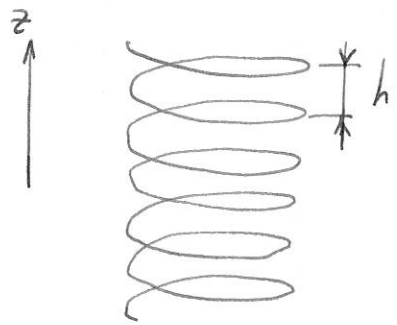
$$z \to z + \alpha \xi$$
$$\phi \to \varphi - \xi$$

which means that

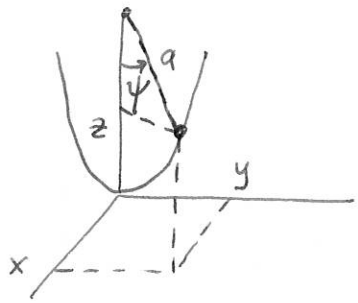
$$\alpha \frac{\partial L}{\partial \dot{z}} - \frac{\partial L}{\partial \dot{\varphi}} = \text{const} \Rightarrow \boxed{\alpha \dot{z} - \rho^2 \dot{\varphi} = \text{const}}$$

Example: potential of an infinite charged (uniformly) coil such that the "step" is equal to

$$h = 2\pi\alpha$$



Problem 2



a) $x = a\sin\psi\cos\omega t$

$y = a\sin\psi\sin\omega t$

$z = a(1-\cos\psi)$

$\dot{x} = a\cos\psi\cos\omega t\,\dot{\psi} - a\sin\psi\sin\omega t\,\omega$

$\dot{y} = a\cos\psi\sin\omega t\,\dot{\psi} + a\sin\psi\cos\omega t\,\omega$

$\dot{z} = a\sin\psi\,\dot{\psi}$

$\dot{x}^2+\dot{y}^2+\dot{z}^2 = a^2\dot{\psi}^2 + a^2\sin^2\psi\,\omega^2$ (cross-terms cancel)

Lagrangian

$$L = \frac{m(\dot{x}^2+\dot{y}^2+\dot{z}^2)}{2} - mgz = \frac{ma^2}{2}\dot{\psi}^2 + \frac{ma^2\omega^2}{2}\sin^2\psi - mga(1-\cos\psi)$$

b) $\ddot{\psi} = -\dfrac{\partial V_{eff}}{\partial\psi}$, where $V_{eff} = \dfrac{g}{a}(1-\cos\psi) - \dfrac{\omega^2}{2}\sin^2\psi$

c) for $\psi \ll 1$

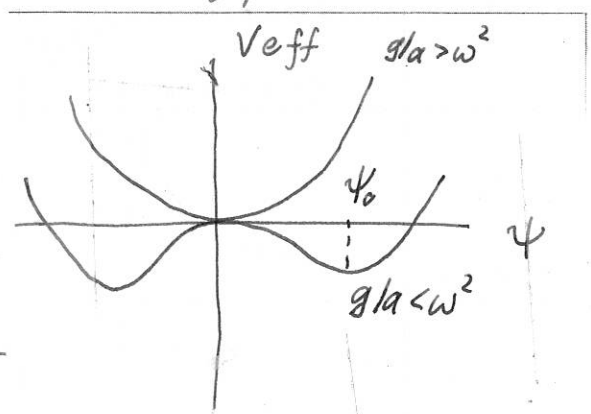
$$V_{eff} \approx \frac{g}{2a}\psi^2 - \frac{\omega^2}{2}\psi^2$$

when $\omega^2 < g/a$, $\psi = 0$ is the position of stable equilibrium.

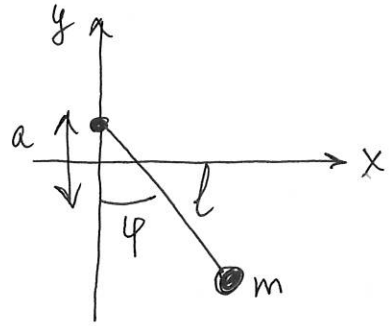
when $\omega^2 > g/a$ the new position is found from

$$\frac{\partial V_{eff}}{\partial\varphi} = 0 \implies \frac{g}{a}\sin\psi_0 - \omega^2\sin\psi_0\cos\psi_0 = 0$$

$$\boxed{\cos\psi_0 = \frac{g/a}{\omega^2} < 1}$$



Pendulum with oscillating support (Kapitza pendulum)



Support point coordinate

$Y(t) = a \cos\gamma t$

Coordinates of m:

$x = l \sin\varphi$

$y = a\cos\gamma t - l\cos\varphi$

$$K = \frac{1}{2} m(\dot{x}^2 + \dot{y}^2) = \frac{1}{2} m \left[(l\cos\varphi\,\dot{\varphi})^2 + (-a\gamma \sin\gamma t + l\sin\varphi\,\dot{\varphi})^2 \right]$$

$$= \frac{1}{2} m \left[l^2\cos^2\varphi\,\dot{\varphi}^2 + l^2\sin^2\varphi\,\dot{\varphi}^2 + a^2\gamma^2\sin^2(\gamma t) - 2la\gamma\sin(\gamma t)\sin(\varphi)\,\dot{\varphi} \right]$$

$$U = mgy$$

$$L = K - U = \frac{1}{2} m \left[l^2\dot{\varphi}^2 - 2la\gamma\sin(\gamma t)\sin\varphi\,\dot{\varphi} \right] + mgl\cos\varphi$$

where I dropped $\frac{1}{2} m a^2\gamma^2 \sin^2(\gamma t)$ and $mga\cos\gamma t$ because these terms do not affect coordinate φ.

Observe that $\sin(\gamma t)\sin\varphi\,\dot{\varphi} = \frac{d}{dt}\left(\sin(\gamma t)\cos\varphi\right) - \gamma\cos(\gamma t)\cos\varphi$

Full time derivative does not affect motion because

$$\int_{t_1}^{t_2} dt \left(L_0 + \frac{dN}{dt} \right) = \int_{t_1}^{t_2} dt\, L_0 + \underbrace{N(t_2) - N(t_1)}_{\text{constant!}}$$

Therefore $L = \frac{1}{2} m \left[l^2 \dot{\varphi}^2 + 2 l a \gamma^2 \cos(\gamma t) \cos\varphi \right] + mgl\cos\varphi$

EL equ: $\frac{d}{dt} \frac{\partial L}{\partial \dot{\varphi}} = \frac{\partial L}{\partial \varphi}$

$$m l^2 \ddot{\varphi} = \underbrace{-mgl\sin\varphi}_{-\frac{dU}{d\varphi}} \underbrace{- m l a \gamma^2 \cos(\gamma t) \sin\varphi}_{f(\varphi, t) = f_1 \cos(\gamma t)}$$

Hence

let now $\varphi = \phi + \xi$

$\Rightarrow$ effective potential (obtained after averaging over fast oscillations with frequency γ)

$$U_{eff}(\phi) = U(\phi) + \frac{f_1^2}{4 m l^2 \gamma^2} =$$

$$= -mgl\cos\phi + \frac{(m l a \gamma^2 \sin\phi)^2}{4 m l^2 \gamma^2}$$

$$= -mgl\cos\phi + \frac{1}{4} m a^2 \gamma^2 \sin^2\phi$$

$$= +mgl\left(-\cos\phi + \frac{a^2\gamma^2}{4gl}\sin^2\phi\right)$$

Minima of $U_{eff}(\phi)$ describe point of stable equilibrium.

Find them: $\frac{dU_{eff}}{d\varphi} = 0 = \frac{d}{d\varphi}\left(-\cos\phi + \lambda \sin^2\phi\right) \quad \left(\lambda = \frac{a^2\gamma^2}{4gl}\right)$

$$\sin\phi + 2\lambda \sin\phi \cos\phi = 0$$

Extrema

(1) $\sin\phi = 0 \Rightarrow \phi = 0$ or π.

(2) $\cos\phi = -\frac{1}{2\lambda} = -\frac{2gl}{a^2\gamma^2}$

Minima: $\dfrac{d^2 U_{eff}}{d\varphi^2} > 0$

$$\frac{d^2 U_{eff}}{d\varphi^2} = \frac{d}{d\varphi}\left(\sin\varphi + \lambda \sin 2\varphi\right) = \cos\varphi + 2\lambda\cos 2\varphi$$

(1) $\varphi = 0 \Rightarrow U''_{eff} = 1 + 2\lambda = 1 + \dfrac{a^2\gamma^2}{2gl} > 0$ <u>YES</u>

$\varphi = \pi \Rightarrow U''_{eff} = -1 + 2\lambda = \dfrac{a^2\gamma^2}{2gl} - 1 > 0$ <u>YES</u>

if $a^2\gamma^2 > 2gl$!

For high frequency γ minimum at $\varphi = \pi$ becomes stable!!!

(2) $\cos\varphi = -\dfrac{1}{2\lambda}$, $U''_{eff} = -\dfrac{1}{2\lambda} + 2\lambda\left(2\cos^2\varphi - 1\right) =$

$$= -\frac{1}{2\lambda} + 2\lambda\left(\frac{2}{(2\lambda)^2} - 1\right) =$$

$$= -\frac{1}{2\lambda} + \frac{1}{\lambda} - 2\lambda = \frac{1}{2\lambda} - 2\lambda =$$

$$= \frac{1 - 4\lambda^2}{2\lambda} > 0 \text{ for } 2\lambda < 1.$$

and $\lambda > 0$.

But $\cos\varphi = \dfrac{-1}{2\lambda}$ requires $|2\lambda| > 1$.

$\Rightarrow$ no minimum at $\cos\varphi = -\dfrac{1}{2\lambda}$.

Back to eq of motion for the slow variable ϕ:

$$ml^2\ddot{\phi} = -\frac{d}{d\phi}U_{eff}(\phi)$$

$$U_{eff} = mgl\left(-\cos\phi + \lambda\sin^2\phi\right)$$

To find period of oscillation we need to expand U_{eff} about minimum ϕ_0 to 2nd order in $\delta\phi = \phi - \phi_0$.

1) let $\phi_0 = 0 \Rightarrow U_{eff} = mgl\left(-(1-\frac{1}{2}(\delta\phi)^2) + \lambda(\delta\phi)^2\right) =$

$$= mgl\left(-1 + \left(\frac{1}{2}+\lambda\right)(\delta\phi)^2\right)$$

$$\Rightarrow ml^2\,\delta\ddot{\phi} = -\frac{d}{d\delta\phi}\left\{-mgl + \frac{1}{2}mgl(1+2\lambda)(\delta\phi)^2\right\}$$

$$= -mgl(1+2\lambda)\,\delta\phi$$

$$\frac{d^2}{dt^2}(\delta\phi) = -\frac{g}{l}(1+2\lambda)\,\delta\phi = -\omega_0^2\,\delta\phi$$

Frequency of oscillation $\omega_0 = \sqrt{\frac{g(1+2\lambda)}{l}}$

Period $T_0 = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{l}{g}}\,\frac{1}{\sqrt{1+\frac{a^2\gamma^2}{2gl}}} < 2\pi\sqrt{\frac{l}{g}}$

2) let $\phi_\pi = \pi \Rightarrow$ write $\phi = \pi + \delta\phi$

$$U_{eff}(\pi+\delta\phi) = mgl\left(-\cos(\pi+\delta\phi) + \lambda\sin^2(\pi+\delta\phi)\right) =$$

$$= mgl\left(\cos(\delta\phi) + \lambda\sin^2(\delta\phi)\right) = mgl\left[1 + \frac{1}{2}(2\lambda-1)(\delta\phi)^2\right]$$

$$\Rightarrow \quad ml^2\,\delta\ddot{\phi} = -\frac{d}{d(\delta\phi)}\left[mgl + \frac{1}{2}mgl(2\lambda-1)(\delta\phi)^2\right]$$

$$= -\,mgl\,(2\lambda-1)\,(\delta\phi)$$

$$\delta\ddot{\phi} = -\omega_\pi^2\,\delta\phi$$

$$\omega_\pi = \frac{g}{l}(2\lambda-1)$$

$$\Rightarrow \quad T_\pi = \frac{2\pi}{\omega_\pi} = 2\pi\sqrt{\frac{l}{g}}\,\frac{1}{\sqrt{2\lambda-1}} = 2\pi\sqrt{\frac{l}{g}}\,\frac{1}{\sqrt{\frac{a^2\gamma^2}{2gl}-1}} > 2\pi\sqrt{\frac{l}{g}}$$

As $2\lambda = \frac{a^2\gamma^2}{2gl}$ approaches 1 from above,

the period of oscillations about $\phi=\pi$ (vertical) position increases. It diverges at $a^2\gamma^2 = 2gl$, which means that the pendulum never returns to $\phi=\pi$ for such a low frequency γ.